

HOW MUCH DO INDIVIDUAL STORES EARN IN A DAY?

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Section 1: The Model

The model that was chosen for this specific assignment, was decided after considering other options, and then going with the one that appeared the most robust yet simple enough that it would not overfit our collected data points

In its simplest definition the model is:

$$\text{revenue} = \text{transactions} \times \text{basket}$$

However, it is clear that each and every basket cannot be measured, neither can each atomic transaction be counted since that would require unfeasible amount of time resource.

So here is what I did instead, I decided to visit each shop and observe for 15 mins, after which I conducted a short interview of whoever was the highest in authority present. During this observation period, I observed the number of transactions and the value of 5 randomly selected transactions out of all of them, amongst other things. I visited these stores at different times to try to ensure a good data distribution, but 15 mins of observation during morning is not equal to peak time, usually night, so I introduced a daypart variable to balance that out.

Below is the final model and the variables that were used for this assignment:

$$R_{\text{daily}} = T_{\text{daily}} \times B_{\text{avg}}$$

$$T_{\text{daily}} = (\text{transactions_in_window} \times (60 / \text{window_minutes})) / \text{daypart_factor} \times \text{operating_hours}$$
$$B_{\text{avg}} = 0.6 \times \text{basket_observed} + 0.4 \times \text{midpoint}(\text{bill_range})$$

$$\text{Cross-check: } R_{\text{inv}} = \sum(\text{restock_per_day} \times \text{qty} \times \text{price}) / 0.6$$

Observed (I saw these):

- transactions_n during 15-min window
- basket_value_obs_1..5 (basket values you heard or saw)
- staff_n, fridges_n, store_size, store_type, location_type, categories_visible
- obs_window_start, obs_window_end, obs_daypart

Asked (owner told):

- open_time, close_time
- bill_range_typical_low, bill_range_typical_high
- top_item_1..3, restock_freq_item_1..3, qty_per_restock_item_1..3, price_item_1..3
- customers_per_day_est_range
- revenue_band_reported (used only for validation, not as input)

Derived (computed):

- daypart_factor (lookup based on obs_daypart)
- daily_txn_est
- B_obs (mean of observed baskets, requires ≥ 3 non-blank)
- B_asked (midpoint of bill range)
- B_avg (blended)
- operating_hours (close - open, with wraparound handling)
- R_low / R_base / R_high
- R_inv
- triangulation_ratio

Other than this, below is the table regarding the daypart factors:

Morning	1.0	(window rate $\approx$ daily average)
Midday	0.9	
Afternoon	0.7	(window rate under-represents the day)
Evening	1.4	(window rate over-represents the day)
Night	0.7	

The **approach for collecting the data** was that I would go as a customer, buy something like chewing gum and then step aside. After that I would start the 15 min observation window, with a timer on my phone and during this window I would not engage with anyone. Once complete, I would approach the owner, explaining that I am a student conducting a research project, to build trust. From there, I had made a google form with all the questions, starting from the easiest and ending on the question regarding daily sales. Instead of having this as a fixed number, I had this as a range since people are more comfortable selecting that than giving out numbers. Out of 10 stores, two refused to tell and one was very hesitant which dropped the confidence scores respectively.

The **model returns bands** because of the following reasons:

- Daypart factors were assumed, Section 4 validates them
- Basket distributions are skewed and a few large bills pull means in unpredictable directions
- The single 15-min window has Poisson variance

Section 5 proposes how these bands can be further narrowed down.

Before proceeding, here are **some honest assumptions**:

- One window's transaction rate, after daypart adjustment, is representative of the day
- The 5 observed basket values approximate the mean (not the mode)
- Owner-reported bill range is roughly symmetric; midpoint is a usable proxy
- Operating hours were limited to 8:00PM indefinitely by the government, which reduced some shops hours by 2-4 hours. Since the other figures were from the last couple of days/weeks, I chose to keep it as is.
- Operating hours are stable day-to-day within ± 1 hour
- Top 3 SKUs cover $\sim 60\%$ of revenue (the inventory model's load-bearing assumption)
- There was no major seasonality during my visit week in Lahore
- Daypart factors (1.0, 0.9, 0.7, 1.4, 0.7) were assumed from convenience-retail intuition, not measured. Section 5 proposes empirical calibration.

Section 2: Justifying the Model

Before proceeding with the current model, I did looking into all the methods that I could have used

Approach	Logic	Why considered	Why rejected (or kept)
A. Flow model ($T \times B$)	Decompose revenue mechanistically	Each input independently checkable	Kept as primary
B. Inventory velocity	Top-SKU movement scaled up	Owners more honest about products than money	Kept as cross-check
C. Feature regression	Fit revenue ~ features	Standard ML approach	Rejected – see below
D. Owner-direct ask	Just ask owner	Easy	Rejected as primary; used only for validation

I did not go with **feature regression** because I had a dataset where $n=10$, meaning there were more candidate features than the degrees of freedom, essentially meaning that real patterns could not be separated reliably from random noise. The dataset was also too small to have a test set, and if I had trained on all of the data then I could not have any independent way to check whether the model generalizes or not. Even if I got any coefficients they would very likely be noise.

The **owner's direct answer alone** fails for multiple reasons, starting from the fact that two of them chose not to answer, outright meaning a 20% data loss. Due to social-desirability, religious ideas and tax reasons under-reporting is very common, and shopkeepers would most likely underquote how much they earn in a day. For KIR01, that was the case since the calculated bands exceeded the owner's claim. This would be a single source bias with no way to correct the bias since there is no other data.

For this case **Fermi decomposition** works since each input can be checked independently, each assumption is noted down and uncertainties can be seen and bounded where they enter. There is no overfitting risk and no test data is needed so that helps.

Finally, I also decided to use **two methods** with **R_inv for validation** since it was independent from the other model using its own inputs, meaning if there was some degree of agreement, the result is more informative. This is basically what allows for triangulation, explained below.

It also allows you to learn something new about the model, such as disagreement between the two could give insights into how some of the variables were reported incorrectly. It also helps bridge the gap between misreporting since shopkeepers tend to provide correct answers more for stock related questions as compared to sales related questions.

Triangulation is just a validation method, where we use the flow model (transactions) and inventory model, both displayed in section 1 and cross check. The **triangulation ratio** is just $R_{inv} \div R_{base}$. A ratio of 1.0 means perfect agreement. You set the acceptable band at 0.7–1.4, meaning the two methods are within 30-40% of each other. Inside the band is "agree", outside is "disagree" and investigated.

One more thing I would like to clarify is that **why I took a weighted average for basket values** between observed values and midpoint of the typical bill values told by the shopkeeper. Observation alone could have some visual/audible bias in some shops where transactions aren't displayed on screens, and the owner's bill value estimates may be incorrect or underreported for the same reasons as revenue is. Weighted blend privileges direct observation (0.6) but doesn't ignore the owner (0.4). In section 4, we will discover more about this parameter.

Section 3: Apply the Model

Store	Type	Size	Daypart	R_low	R_base	R_high	R_inv	Owner band	Triang.	Confidence
MART03	mini mart	Large	Evening	33k	265k	825k	148k	260-280k ✓	0.56 △	9
MART01	mini mart	Medium	Evening + Midday	14k	227k*	704k	102k	refused	0.45 △	6
KIRO3	kiryana	Large	Afternoon	21k	214k	686k	207k	200-220k ✓	0.97 ✓	8
KIRO2	kiryana	Medium	Night	31k	162k	514k	220k	160-180k ✓	1.36 ✓	9
KIRO1	kiryana	Medium	Evening	12k	123k	393k	119k	80-100k ✗	0.97 ✓	7
MART02	mini mart	Medium	Afternoon	18k	101k	300k	93k	refused	0.92 ✓	5
MART04	mini mart	Large	Midday	23k	87k	192k	83k	80-100k ✓	0.95 ✓	9
PAN01	pan shop	Small	Evening	5k	64k	209k	77k	60-80k ✓	1.19 ✓	9
PAN03	pan shop	Small	Evening	10k	43k	99k	56k	40-60k ✓	1.30 ✓	9
PAN02	pan shop	Small	Morning	4k	33k	75k	40k	20-40k ✓	1.22 ✓	9

✓/△ symbols: ✓ for owner band contains base / triangulation in [0.7, 1.4], △ otherwise.

*MART01 R_base is the mean of two visits (233k Evening, 221k Midday)

Starting off with the **wide bands**, the model uses the shop keeper's reported lower and higher end of bills normally charged, and since the observation window is just 15 mins the band is what that window can defend, not what the model believes most likely.

Then, out of the 8 **shopkeepers that reported figures** for the daily sales, the flow model (transactions x basket) agreed with 7 of them, which validates the model's capability to a certain extent. The one where it did not was **KIRO1** (123k vs 80-100k reported), where the confidence score was 7/10 because the shopkeeper hesitated to tell the sales figures, most likely underreporting them.

Moving on we have **8/10 stores "agree" in traingulation ratio**, suggesting the flow model and inventory model estimated were within 30-40% of each other which has been set as the borderline acceptable number. However, MART01 and MART03, have flows much greater than inventory. This suggests that the marts probably have lower top-SKU concentration than the assumed 60%, which is plausible since these are mini marts with wider product mix and top 3 cover less. This is to be addressed in section 5.

Lastly, while data is not representative I tried to cover **different geographic areas** DHA, Gulberg, Johar Town, Ichhra, Model Town and went during **different time windows**, despite most being in evening. This is the sort of process where improvement can be made as sampling will be addressed in section 5.

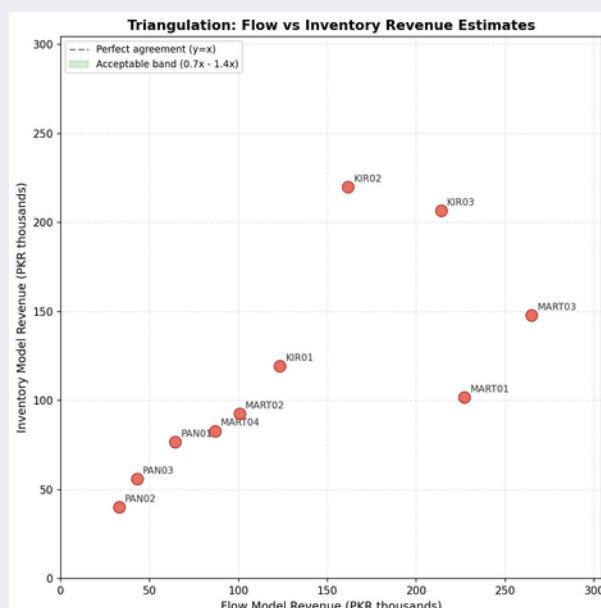
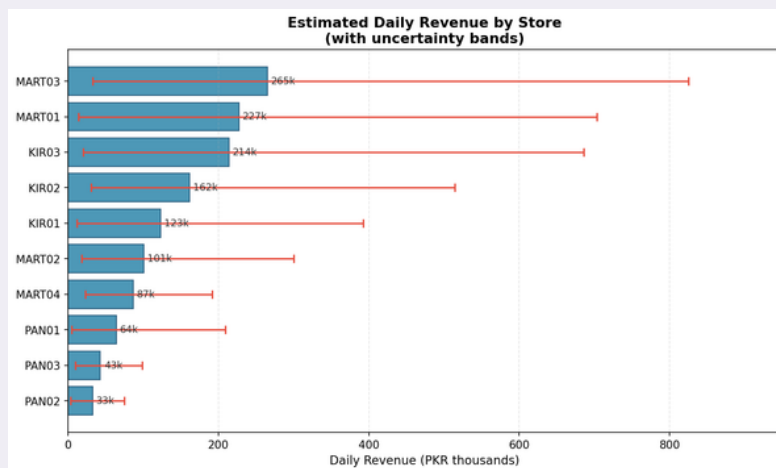


Chart 1 shows that the estimated revenues follow a realistic pattern: marts and larger kiriyana stores appear at the top, while pan shops have the lowest estimated daily revenues. This suggests that the model is producing sensible rankings rather than random results. However, the uncertainty bands are very wide, especially for higher-revenue stores like MART03 and MART01. This happens because the bands are based on best- and worst-case assumptions, so larger stores naturally get larger absolute ranges. The upper bands are also much wider than the lower bands because owner-reported bill sizes can have very high upper limits.

Chart 2 compares two separate ways of estimating revenue: the customer-flow method and the inventory-based method. Most stores are reasonably close to the agreement line, which supports the overall direction of the estimates. Pan shops cluster tightly at the lower end, suggesting their estimates are more stable because their product mix is simpler. The main exceptions are MART01 and MART03, where the flow-based estimate is much higher than the inventory estimate. This suggests that the inventory method may understate mini-mart revenue, possibly because the “top SKUs cover 60% of sales” assumption works better for small shops than for marts with broader product ranges.

Section 4: Validation

Standard accuracy metrics would not have worked for this specific case. For validation purposes, I mainly used these tests:

- V1: agreement with owner-reported bands
- V2: agreement between two independent estimation methods (flow vs inventory)
- V3: internal consistency across two visits to one store (MART01)
- V4: sensitivity to assumed parameters

V1 – Owner band coverage

This test simply checks if the R_{base} value lands in the range the shop owner had selected or not.

Result: 7/8 shops pass this test

Store	R_base	Owner band	Contains?
KIR01	123k	80-100k	✗ (above)
KIR02	162k	160-180k	✓
KIR03	214k	200-220k	✓
MART03	265k	260-280k	✓
MART04	87k	80-100k	✓
PAN01	64k	60-80k	✓
PAN02	33k	20-40k	✓
PAN03	43k	40-60k	✓

This was just a soft test. Owner bands are themselves biased and in informal retail underclaims for tax/social reasons are very common. There was a miss where $R_base > \text{band}$ in KIR01, most likely due to this reason. 2 stores (MART01, MART02) refused entirely and were not assessed in this.

V2 – Triangulation gap

This test calculated **Median $|R_inv - R_base| / R_base$**

Result: across 10 stores = ~21%

The result shows an acceptable amount of median gap in triangulation, since our earlier accepted ratios results were bounded between 0.7-1.4, a **30-40% gap was deemed acceptable**

The issue as discussed earlier was with MART01 and MART03, both exceeding the acceptable gap. The reason most likely lies in top-SKU share assumption (60%) breaking for marts where SKU mix is broader, a hypothesis. Section 5 will address this.

Interestingly, MART01 disagreed in both visits as presented by the dataset attached with this report, indicating that this was not noise

Store	Gap
KIR01	3%
KIR03	3%
MART04	5%
MART02	8%
PAN01	19%
PAN02	22%
PAN03	30%
KIR02	36%
MART03	44%
MART01	55%

V3 — Two-window consistency

This test was done between two visits to MART01 on different days and timing.

Result: R_base = 233k for visit 1, 221k for visit 2. Gap = 4.98% of mean.

This is important validation, because there is no clearly visible bias present. Two independent observations at different dayparts (Evening + Midday) produced near-identical estimates. **This suggests that the daypart correction is doing real work**, and is not just working on luck. That being said, this should be done for multiple stores, not just one, as will be propose in section 5

V4 — Sensitivity Analysis

I wanted to test the values that I had assumed, so I decided to do a test by increasing/decreasing them to see how the results change. Below is teh summary of the changes and their impact on the model.

Perturbation	Median ΔR_{base}	Max ΔR_{base}	Triangulation flips
Daypart compressed	6.10%	7.9% (KIRO2)	1
Daypart steepened	5.40%	9.4% (KIRO3)	0
Top-SKU share 0.50	— (R_inv only)	—	4
Top-SKU share 0.70	— (R_inv only)	—	0
Basket weight 0.4/0.6	10.40%	13.3% (KIRO2)	0
Basket weight 0.8/0.2	10.40%	13.3% (KIRO2)	1

The results suggested the following:

- **Daypart factors matter less than expected.** $\pm 20\%$ perturbation produces only $\sim 6\%$ R_base swing. The band width already absorbs this.
- **Basket weight matters more.** Median 10% swing on R_base. KIRO2 is most affected because B_obs (288) and B_asked (550) diverge most for that store.
- At top-SKU share 0.50, 4 stores' triangulation status flips from agree to disagree. At baseline 0.60, the marginal stores (KIRO2, PAN01, PAN02, PAN03) sit between 1.19 and 1.36. This means the V2 pass rate depends on assumption of 60% or more of top 3 item share being correct

However, it is **also important to note that PAN shops (3/4 flagged) do get most of their sales from the top 3 items that our dataset shows**. This needs to be investigated further as this seemed to be a weakness in the model.

Section 5: What you'd do differently

here is everything I would do:

Bigger Sample

Issue: The current sample is small, purposive, and geographically clustered in Lahore, so it cannot support population-level claims.

Solution: I would use a stratified random sample of 200–500 stores. The strata would include store type, neighborhood income level, and location type such as residential areas, markets, main roads, transit points, and school areas. If possible, I would also include 2–3 cities to test whether a Lahore-based model generalizes outside Lahore.

Why: This would make the estimates more representative and allow broader claims about small retail revenue patterns.

Repeat Visits

Issue: Single visits do not capture day-of-week effects, weather variation, or natural fluctuations in customer traffic. The current repeat-visit evidence is weak because only one store was revisited.

Solution: Each store would be visited three times at different dayparts and on different days. The 15-minute observation window could be retained, but the repeated visits would allow the model to average across observations.

Why: This would replace assumed traffic bands with empirical variation. Instead of saying revenue may vary by $\pm 25\%$, the model could use actual observed variation from each store.

Calibrating Key Assumptions

Issue: Several important parameters are currently assumed rather than measured, especially daypart factors, top-SKU share, and basket-weight blending.

Solution: I would select 8–10 anchor stores for full-day observation to measure actual transaction patterns across the day. This would allow the model to calculate real daypart factors instead of relying on convenience-retail intuition. For top-SKU share, the truth subset would be used to measure how much revenue comes from the top three SKUs by store type. Pan shops, kiryanas, and marts should not all use the same 60% assumption. For basket weight, electronic transaction capture at selected stores would be compared with audible observations and owner-stated bill ranges to derive better store-type-specific weights.

Why: These assumptions directly affect revenue estimates. Calibrating them would reduce error, especially for marts where the current triangulation issues may be caused by lower top-SKU concentration.

Truth Subset

Issue: The biggest limitation is the lack of true daily revenue, so the model's absolute accuracy cannot be proven.

Solution: Around 10–15% of the sample, roughly 25–50 stores, would be selected as a truth subset. Two enumerators would observe each of these stores for the full operating day and record every transaction and basket value.

Why: This would create a reliable validation set. The truth subset could be used to calibrate daypart factors, top-SKU shares, basket weights, and model accuracy.

Use of the Larger Dataset

Issue: With only 10 stores, feature regression or machine learning would be inappropriate.

Solution: With 200–500 stores and a truth subset, the Fermi model could become the starting label, while the truth subset would provide high-confidence labels. A supervised model such as gradient-boosted regression could then predict daily revenue using observable features like shelf meters, SKU breadth, fridges, staff count, location score, operating hours, and neighborhood income.

Why: The Fermi model should come first because it is interpretable and checkable. Machine learning should come second, adding scale and predictive power while still being tested against the Fermi estimates and truth subset.

Quality Control

Issue: A single-enumerator setup gives no inter-rater reliability check. Data-entry delays and inconsistencies also create risk.

Solution: Collectors would receive two days of training before fieldwork. On a calibration day, all collectors would observe the same 2–3 stores and compare their counts. Data entry would happen daily, with same-day field notes. Weekly audits would check for missing fields, outliers, suspiciously round numbers, and inconsistencies. In the truth subset, two enumerators would independently observe the same store to measure reliability.

Why: This would make the data more consistent, reduce manual cleaning issues, and make the fieldwork defensible.

Appendix

Google form used:

https://docs.google.com/forms/d/e/1FAIpQLSdl_B8iVqaCxTGZBgQ6SjttOSSrLDyXlfvBQwbC5V_e2JDfhg/viewform